CS2580

UNIVERSITY OF WARWICK
LEVEL 5 Open Book Assessment [2 hours]
Department of Computer Science
CS2580 DATABASE SYSTEMS

Instructions
1. Read all instructions carefully – and read through the entire paper at least once before you start writing.
2. There are 5 questions. You should **attempt [4] questions.** You should not submit answers to more than the required number of questions
3. All questions will carry the same number of marks unless otherwise stated.
4. You should handwrite your answers either with paper and pen or using an electronic device with a stylus (unless you have special arrangements for exams which allow the use of a computer). Start each question on a new page and clearly mark each page with the page number, your student id and the question number. Handwritten notes must be scanned or photographed and all individual solutions should (if you possibly can) be collated into a single PDF with pages in the correct order. You must upload two files to the AEP: your PDF of solutions and a completed cover sheet. You must click **FINALISE** to complete the submission process. After that you have done so you will not be able to upload anything further.
5. Please ensure that all your handwritten answers are permitted written legibly, preferably in dark blue or black ink. If you use a pencil ensure that it is not too faint to be captured by a scan or photograph.
6. Please check the legibility of your final submission before uploading. It is your responsibility to ensure that your work can be read.
7. You are allowed to access module materials, notes, resources, references and the internet during the assessment.
8. You should not try to communicate with any other candidate during the assessment period or seek assistance from anyone else in completing your answers. The Computer Science Department expects the conduct of all students taking this assessment to conform to the stated requirements. Measures will be in operation to check for possible misconduct. These will include the use of similarity detection tools and the right to require live interviews with selected students following the assessment.
9. By starting this assessment you are declaring yourself fit to undertake it. You are expected to make a reasonable attempt at the assessment by answering the questions in the paper.

CS2580

Please note that:

- You must have completed and uploaded your assessment before the 24 hour assessment window closes.
- You have an additional 45 minutes beyond the stated length of the paper to allow for downloading and uploading the assessment, your files and technical delays.
- For further details you should refer to the AEP documentation.

Use the AEP to seek advice immediately if during the assessment period:

- you cannot access the online assessment;
- you believe you have been given access to the wrong online assessment.

Please note that technical support is only available between 9AM and 5PM (BST).
Invigilator support will be also be available (via the AEP) between 9AM and 5PM (BST).

Notify comp-sci@dcs.warwick.ac.uk as soon as possible if you cannot complete your assessment because:

- you lose your internet connection;
- your device fails;
- you become unwell and are unable to continue;
- you are affected by circumstances beyond your control (e.g. fire alarm).

Please note that this is for notification purposes, it is not a help line.

Your assessment starts below.

1. Consider a database for hotel bookings:

   HOTEL (hotelNo, hotelName, city)
   ROOM (roomNo, hotelNo, type, price)
   BOOKING (hotelNo, guestNo, dateFrom, dateTo, roomNo)
   GUEST (guestNo, guestName, guestAddress)

   You must use the aliases H, R, B, G, respectively, for the table names in the queries below.
   Assume that guestAddress is of type VARCHAR[50], containing the city name.
   Similarly, guestName is of type VARCHAR[30], containing the first name.

   Specify the following queries in SQL:

   (a) List the names and addresses of all guests living in London, with a booking in a hotel in 'London', alphabetically ordered by name. You must use explicit joins (i.e., using the correct 'JOIN' operator).
   [4]

   (b) List the guest numbers associated with the first name 'Peter' having made bookings with unknown dateTo, using implicit joins (i.e., without 'JOIN' operators but with FROM clauses with more than 1 table).
   [4]

   (c) Repeat (b) above but without using any explicit or implicit joins.
   [8]

   (d) List the price and room number of rooms at the Hilton Hotel. Additionally, the room is currently occupied list also the name of the guest staying in the room. You may use the CURDATE() function and assume that all dates are strings of the form yyyymmdd.
   [9]

   Total mark for this question: 25

(continued)

2.

a. Relations R(A, B) and S(B, C) contain $N_R > 0$ and $N_S > 0$ tuples, respectively. For each of the following relational algebra expressions, state in terms of N_R and N_S the minimum and maximum number of tuples that could be in the result of the expression, while making no assumptions about keys:

i. $R \cup (\rho_{S(A, B)} S)$

ii. $\pi_{A,C} (R \bowtie S)$

iii. $\pi_B R - (\pi_B R - \pi_B S)$

iv. $(R \bowtie R) \bowtie R$

v. $(\sigma_{A>B} R) \cup (\sigma_{A<B} R)$

[10]

b. In relations R(A, B) and S(B, C) all attribute domains are integers. Making no assumptions about keys, consider the following three relational algebra expressions. Two of them are equivalent (i.e., produce the same answer on all databases), while one of them can produce a different answer. Find the query that can produce a different answer? Provide the simplest DB state where a different answer is produced.

i. $\pi_{A,C} (R \bowtie \sigma_{B=1} S)$

ii. $\pi_A (\sigma_{B=1} R) \times \pi_C (\sigma_{B=1} S)$

iii. $\pi_{A,C} (\pi_A R \times \sigma_{B=1} S)$

[8]

c. Relation RT(regionID, name, highT, lowT) has regionID as the key. Relation RT stores historical records of high and low temperatures for various regions. With just one sentence describe what will be the output of executing the following query:

T1(rID, h) = $\pi_{regionID, highT} (RT)$

T2(rID, h) = $\pi_{regionID, lowT} (RT)$

T3(regionID) = $\pi_{rID} (T2 \bowtie_{h<highT} RT)$

T4(regionID) = $\pi_{rID} (T2 \bowtie_{l>lowT} RT)$

T5(regionID) = $\pi_{regionID} (RT)$ — T3

T6(regionID) = $\pi_{regionID} (RT)$ — T4

Result(n) = $\pi_{name} (RT \bowtie (T5 \cup T6))$

[7]

Total mark for this question: 25

(continued)

3. Consider the COMPANY database schema shown below and provide the following SQL queries.

EMPLOYEE

Fname	Minit	Lname	Ssn	Bdate	Address	Sex	Salary	Super_ssn	Dno

DEPARTMENT

Dname	Dnumber	Mgr_ssn	Mgr_start_date

DEPT_LOCATIONS

Dnumber	Dlocation

PROJECT

Pname	Pnumber	Plocation	Dnum

WORKS_ON

Essn	Pno	Hours

a. Retrieve the Social Security Number of the supervisor(s) whose employees (supervisees) have an average salary greater than the average salary among all employees who work for a department in 'Leamington'. You must use at most 1 (two-way) join and one GROUP BY.

[6]

b. Retrieve the salaries of all employees whose supervisor's supervisor has a salary that is not the minimum among all supervisor salaries. Your solution must use the ANY operator.

[6]

c. Retrieve the names and salaries of employees living outside London whose salaries exceed by more than 20% the minimum salary of employees who are living in London and are working on project 'X'. You are not allowed to use rownum, TOP, or LIMIT.

[5]

d. Retrieve the 3rd maximum salary among all employees. You must not use any joins.

[8]

Total mark for this question: 25

(continued)

4.

a. Consider the relation shown below.

X	Y	Z
x_1	y_1	z_1
x_1	y_1	z_2
x_2	y_1	z_1
x_2	y_1	z_3

i. List all nontrivial functional dependencies that this relation instance satisfies.

[3]

ii. Assume that the value of attribute Z of the last record in the relation is changed from z_3 to z_2. Now list all the functional dependencies that this relation instance satisfies.

[2]

b. Consider the following collection of relations and dependencies per each relation. [Assume that each relation (R1, …, R5) is obtained through decomposition from a relation with attributes ABCDEFGHI and that all the known dependencies over relation ABCDEFGHI are listed below for each relation.]

- R1(A,C,B,D,E), A → B, C → D
- R2(A,B,F), AC → E, B → F
- R3(A,D),D→G,G→H
- R4(D,C,H,G), A → I, I → A
- R5(A,I,C,E)

use a single sentence to:

i. State the strongest normal form each one of the relations (R1, …, R5) is in.
ii. If it is not in BCNF, decompose it into a collection of BCNF relations.

[5]

c. Consider the relation schema R(A,B,C), which has the FD B → C. If A is a candidate key for R, is it possible for R to be in BCNF? If so, under what conditions? If not, explain why not.

[2]

d. Consider a relation *R(A,B,C,D,E)* with the following dependencies:

AB -> C *CD -> E* *DE -> B*

Is *AB* a candidate key of this relation? If not, is *ABD*? Explain your answer with one sentence.

[3]

e. Consider the relation R, which has attributes that hold schedules of courses and sections at a university; R = {CourseNo, SecNo, OfferingDept, CreditHours, CourseLevel, InstructorSSN, Semester, Year, Days_Hours, RoomNo, NoOfStudents}. To simplify, let us use the following shorthand notation: C = CourseNo, SN = SecNo, OD = OfferingDept, CH = CreditHours, CL = CourseLevel, I = InstructorSSN, S = Semester, Y = Year, D = Days_Hours, RM = RoomNo, NS = NoOfStudents. Hence, R = {C, SN, OD, CH, CL, I, S, Y, D, RM, NS}.
The following functional dependencies hold:

{C} -> {OD, CH, CL}
{C, SN, S, Y} -> {D, RM, NS, I}
{RM, D, S, Y} -> {I, C, SN}

Determine the candidate keys of R and write it in one sentence per key. Normalize R to BCNF, writing one sentence per decomposed relation schema and the highest NF it satisfies.

[10]

Total mark for this question: 25

(continued)

5.

a. Consider a disk with a sector size of 512 bytes, 2000 tracks per surface, 50 sectors per track, five double-sided platters, average seek time of 10 msec, and its platters rotate at 5400 revolutions per minute.
   i. Is 256 bytes a valid block size? How about 2048? How about 51200?
   [2]
   ii. What is the average rotational delay?
   [2]
   iii. If one track of data can be transferred per revolution, what is the disk's transfer rate?
   [6]

b. Assume that for the above disk a block size of 1024 bytes is chosen. Suppose that a file containing 100,000 records of 100 bytes each is to be stored on this disk and that no record is allowed to span two blocks.
   i. How much time is required to read this file (into memory) if it was stored sequentially on disk (contiguous allocation)?
   [3]
   ii. How much time is required to read the file (into memory) in an indexed allocation, when the index is in memory? Assume memory costs are negligible.
   [4]
   iii. How much time is required to read a single random block of the file (into memory) assuming the file is sorted on disk?
   [8]

Total mark for this question: 25

(END)